

CURRICULUM VITAE

Dr.Ajijur Rehman

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PROFESSIONAL SUMMARY

Accomplished Bioinformatics Researcher and Educator with a Ph.D. in Bioinformatics and 7 years of teaching and research experience. Specialized in **Computer-Aided Drug Design, Nanoinformatics, and Health Informatics**, with a proven ability to integrate advanced computational techniques into biomedical and healthcare research. Adept at mentoring students and leading interdisciplinary research initiatives aimed at pioneering breakthroughs in drug discovery and therapeutic innovations. Passionate about contributing to the future of healthcare through cutting-edge research and teaching.

EDUCATIONAL QUALIFICATIONS

Ph.D. in Bioinformatics | Integral University, Lucknow, India | **2019**

M.Sc. in Bioinformatics | Integral University, Lucknow, India | **2008**

B.Sc. (Zoology, Botany, Chemistry) | University of Lucknow, India | **2005**

Ph.D THESIS TITLE: Design and optimization of leads for novel drug targets against *Mycobacterium Tuberculosis*

ACADEMIC EXPERIENCE

Assistant Professor

NIMS Institute of Allied Medical Science and Technology, NIMS University, Jaipur, Rajasthan | *October 2021 – June 2024*

- Designed and delivered courses on Bioinformatics, Computational Biology, and related subjects.
- Supervised undergraduate and postgraduate research projects in drug design and informatics.
- Conducted interdisciplinary research utilizing computational methodologies to solve biomedical challenges.

Assistant Professor

Department of Biotechnology, R.R. Institute of Management and Technology, Lucknow, India | *September 2019 – September 2021*

- Facilitated courses on Bioinformatics, Biotechnology, and related disciplines.
- Led student research projects integrating bioinformatics tools for drug discovery and molecular analysis.

Assistant Professor

Department of Biotechnology, Dr. M.C. Saxena Group of Colleges, Lucknow, India | *August 2017 – August 2019*

- Instructed courses in Bioinformatics and Biotechnology.

- Developed curriculum and educational material to enhance student learning in biomedical and computational sciences.

EXTRA-CURRICULAR AND ADMINISTRATIVE ROLES

- **NAAC Criteria II Coordinator**
- Led the NAAC Criteria II preparation and documentation for accreditation, focusing on Teaching-Learning and Evaluation.
- Analysed and improved teaching methodologies, student feedback systems, and evaluation mechanisms.
- **Departmental Exam Coordinator**
- Coordinated all departmental exams, ensuring that exam schedules, communication, and logistics ran smoothly.
- Worked with faculty and administration to ensure timely and accurate assessments of student performance.
- **Event Coordinator**
Organized departmental events, including seminars, workshops, and conferences.
- Managed the planning, execution, and coordination of events to ensure seamless operations.
- **Sports Coordinator**
- Organized sports events to promote physical well-being among students.
- Coordinated inter-departmental and inter-college sports competitions, ensuring active participation and smooth conduct.
- **Sustainable Development Goal (SDG) II – Zero Hunger**
- Led initiatives focused on combating hunger and promoting food security through local food banks and awareness campaigns.
- Collaborated with local organizations to advocate for sustainable agricultural practices and nutrition.
- **Curriculum Development Coordinator**
- Contributed to syllabus revision and program accreditation efforts.
- Worked with faculty to enhance course offerings and instructional methods.
- **Postgraduate (PG) Student Mentor**
- Supervised PG students in research projects, dissertations, and thesis writing.
- Advised students on bioinformatics research methodologies and computational tools.
- Assisted students in publishing research papers and presenting at conferences.

Research Paper published in Scopus/National/International Journal

- **Rehman Ajijur** Khan Mohammad Kalim Ahmad, **Exploring Small Molecules as Novel Drug Candidates Targeting 4-Hydroxy-tetrahydronicotinate Reductase in Mycobacterium tuberculosis: Insights from Molecular Dynamics Simulation and Principal Component Analysis**, *Latter's in Applied NanoBioscience*. (2024) September 21, Volume 14, Issue 1, 2025,2
- **Rehman A**, Akhtar S, and Khan MKA. **Combinatorial design to Decipher Novel Lead Molecule against Mycobacterium tuberculosis**. *Biointerface Research in Applied Chemistry* (2021), Feb 7, 11(5): 12993-13004.
- **Rehman A**, Akhtar S, Siddiqui MH and Khan MKA. **In Silico Identification of Drug-Like Inhibitors against Mtb-DHPS: A Shape-Based Approach**. *J. Pharm. Sci. & Res.*(2019), 11(5): 2084-2089.

- **Rehman A**, Akhtar S, Siddiqui MH, Sayeed U, Ahmad SS, Arif JM, Khan MK. **Identification of potential leads against 4-hydroxytetrahydrodipicolinate synthase from *Mycobacterium tuberculosis***. Bioinformation (2016) Dec 2, 12(11): 400-406.
- Verma A, Mohd. S, Rizvi D, Shaikh S, Ansari M A, Shakil S, Ghazal F, Siddiqui M H, Haneef M, and **Rehman A** **"Compounds isolated from *Ageratum houstonianum* inhibit the activity of matrix metalloproteinases (MMP-2 and MMP-9): An oncoinformatics study"**. Pharmacogn Mag. 2014 Jan-Mar; 10(37): 18–26.

Research Paper Published in UGC Care/Peer Reviewed Journal

- Noorul Amin¹, **Dr Ajijur Rehman²**, **Exploring Novel Strategies for Androgen Receptor Activity: via Computational Biology a Therapeutic Interventions in Androgen-Driven Disease (PHASE 1)**, INTERNATIONAL JOURNAL OF RESEARCH AND ANALYTICAL REVIEWS (IJRAR) | IJRAR.ORG An International Open Access, Peer-reviewed, Refereed Journal, 2024 IJRAR JUNE 2024, Volume 11, issue 2
- Ishan Joshi¹, **Dr. Ajijur Rehman²**, **In Silico Exploration of Hsp90 as a Therapeutic Target in Skin Cancer: Computational Perspectives**, International Journal for Multidisciplinary Research (IJFMR) E-ISSN: 2582-2160, Volume 6, issue 3 May-June 2024.
- Harshita Seth¹, **Dr. Ajijur Rehman²**, **Computational Exploration and Molecular Docking Analysis of *Mycobacterium Tuberculosis* RPF Protein**, International Journal for Multidisciplinary Research (IJFMR) E-ISSN: 2582-2160, Volume 6, issue 3 May-June 2024.
- Surya S Krishnan¹, **Dr. Ajijur Rehman²**, **In-Silico Screening of Novel Drug Against Ovarian Cancer by Phytochemicals from *Carica Papaya***, International Journal for Multidisciplinary Research (IJFMR), E-ISSN: 2582-2160- Volume 6, issue 3 May-June 2024.
- Aakash, Ryali Shri Vidya, **Dr.Ajijur Rehman**. **Protein Structure Prediction and Examination of Nucleocapsid Protein from SARS-CoV2 using Computational and Simulation Tools**. International Journal of Science and Research (IJSR) 12 (2023/4/4), 799-817

Research Paper Published in International Conference

- **Ajijur Rehman¹** Mihir Deshnehare², Goldi Tiwar³, Nagendra Kumar Pradhan⁴, Ainyo YL⁵, Mohammad Kalim Ahmad Khan⁶. **Exploring Target-Based Screening, Molecular Dynamics Simulation and Principal Component Analysis for Drug Repurposing in NUT Midline Carcinoma**, INTERNATIONAL SCIENTIFIC AND PRACTICAL CONFERENCE "STATUS AND DEVELOPMENT PROSPECTS OF FUNDAMENTAL AND APPLIED MICROBIOLOGY: THE VIEWPOINT OF YOUNG SCIENTISTS" 25-26 September, 2024

BOOK CHAPTERPUBLISHED

- Muteeb G, Mohd. S, Rizvi D, Shakil S, Alam A, Tripathi R, Ghazal F, **Rehman A, Role of Matrix Metalloproteinases in Cancer. In “Advances In Protein Chemistry”**, Ashraf GM & Sheikh IA (Eds.). OMICS. 2013, 1-19.

PAPER PRESENTATION

- Paper present on the Topic "**Identification of Common Target Proteins For Neurodegenerative Disorders Through Metabolic Pathway Approach** " in **16th Indian Agricultural Scientists& Farmers Congress** on Nanobiotechnological Approaches For Sustainable Agriculture & Rural Development on **22-23 February, 2014** at INTEGRAL UNIVERSITY, Lucknow under the auspices of Integral University, Lucknow and Bioved Research Institute of Agriculture and Technology, Allahabad.

ABSTRACTS PUBLISHED

- Participated and presented a paper entitled "**Proteome retrieval of Mycobacterium tuberculosis H₃₇R_v using bioinformatics tools**" in the seminar on "Scientific tribute to country's first education minister honorable Maulana Abul Kalam Azad" held on 8th November, 2012 at Department of Biotechnology, Integral University Lucknow.

PATENTS

- Pharmaceutical Evaluation of Methanolic Extract of Phyllanthus Niruri-based Polymeric Nanoparticles for the Management of Diabetes Mellitus. Patent filed on April 9, 2023.
- Antifertility Potential of Calendula Officinalis in Male Rats Model. Patent filed on September 22, 2023.

WORKSHOP, SYMPOSIUM and CONFRENCES

- Participated in One day workshop on "**Workshop on Intellectual Property Rights**" held on 25 November 2023. Organized by Department of Medical Laboratory Technology, School of Allied Health sciences and Institute's Innovation Council, Galgotiya University Greater NOIDA, UP
- Participated in "**Two weeks Faculty Development Programme on Entrepreneurship (FEDP)**" held from 03 to 17 August 2023, organized by the Institute's Innovation Council (IIC), School of Biomedical Sciences and School of Business, Galgotiya University, Greater NOIDA, UP
- Participated in two days International Conference on "**Biotrendz Advancement and Challenges in Science and Technology**" organized by NIET Greater NOIDA 24-03-2023 to 25-03-23.
- Participated in as a sports coordinator in the one week sports program '**Spardha 2022**' Organized by Nims University, Jaipur Rajasthan 02-02-2022 to 05-03-2022
- Two weeks Professional Development Programme on "**New Trends and Advancements in Bioengineering**" Organized by Department of Bioengineering, Integral University, Lucknow 15-11-2021 to 29-11-2021

- One week online Faculty Development Program on “**Data Management**” from 18th-22th January 2021 sponsored by Dr. A.P.J. Abdul Kalam Technical University, Lucknow and organized by R.R. Institute of Modern Technology, Lucknow.
- Participated in two days’ workshop on “**PYTHON**” and completed a mini project at Integral University, 18th-19th January 2017.
- Participated in three days International Congress on **Post-Harvest Technologies of Agricultural Produce for Sustainable Food and Nutritional Security** on 10th -12th November, 2016 Organized by the Department of Bioengineering and Biosciences, Integral University, Lucknow.
- Participated in International Level Conference on **Biological advancements in Free Radical Biology and Medicine (ICBAFM)**" on 14th – 16th Nov., 2015 Organized by the Department of Bioengineering and Biosciences, Integral University, Lucknow.
- Participated in the National Level Conference on "**Nanosciences, Nanotoxicology and Nanoinformatics**" Present and future perspectives on 14th – 15th March, 2015. Organized by the Department of Bioengineering and Biosciences, Integral University, Lucknow.
- Participated in the National Level Conference on "**Nanoscience and Biotechnology – Present & Future Prospectives**" organized by Department of Nanobiotechnology, Life Science Foundation India, Karnataka in collaboration with Department of Bioengineering, Faculty of Engineering, Integral University, Lucknow on 20th April 2013.
- Participated in the National Level Conference on "**Fundamentals of nanoscience and biotechnology**" organized by Department of Nanobiotechnology, Life Science Foundation India, Karnataka in collaboration with Department of Bioengineering, Faculty of Engineering, Integral University, Lucknow on 28-30 January 2013.

RESEARCH SKILLS & TOOLS

- **Bioinformatics Tools:** AutoDock, Modeler, Swiss PDB Viewer, ClustalW, MolGro, PILEUP, EMBOSS
- **Databases:** NCBI, Swiss-Prot, PDB, KEGG, PUBMED, Super Natural Database
- **Programming Languages:** PERL, Python(limited use)
- **Operating Systems:** Windows, Linux
- **Software:** MS Office (Word, Excel, PowerPoint), Photoshop, CorelDraw

COURSES TAUGHT

- Molecular Modeling and Drug Design
- Genomics, Proteomics, and Pharmacogenomics
- Applied Bioinformatics
- Molecular Biology, Structural Biology, Biochemistry
- Bioinformatics in Agriculture and Biotechnology
- Genetics Recourse and IPRs
- Bioethics, Biosafety and IPRs

DECLARATION

I, Dr. Ajijur Rehman, hereby declare that the details provided above are true to the best of my knowledge and belief.

Place: Jaipur

Date:

(Ajijur Rehman)

References

1.Dr. Sonalika Singh

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